

COM410 Programming in Practice

A5. Stacks and Queues

CHALLENGE

Snap!

Implement the class **Snap** to play the classic card game according to the following rules.

- The application should generate a shuffled deck of 52 cards and display them in shuffled order.
- The rank of each consecutive pair of cards is compared and the message “SNAP!!!” is displayed when a match is found (i.e., for 52 cards, 51 comparisons are made).
- After all comparisons are made, the application reports the number of snaps found.

Extend the **Snap** class to check for **supersnaps** (where the cards are of the same colour as well as the same rank). Add new methods to the **Card** class to help you as required.

- The message “SUPERSNAP!!!” should be displayed when a snap with similarly coloured cards is found.
- The final output should report both the number of supersnaps and the number of snaps.

Please upload a screenshot of your solution running to the A5 Challenge submission link on Blackboard, by the deadline shown there.